

ELK - Extended Logic Keys

Third-party integration reference: hotkey slots for AtmosphereAutopilot, MechJeb2 and NavUtilities

ELK version **1.1.0** for KSP 1.12.5. Source, releases and README: <https://github.com/Rjoande/ELK>. License MIT. Document date 2026-09-24.

Supported mods and the versions this document was verified against

AtmosphereAutopilot 1.6.1 by Boris-Barboris: <https://github.com/Boris-Barboris/AtmosphereAutopilot>. The actions marked "fork" exist as native hotkeys only in the GA-hotkeys fork (<https://github.com/Rjoande/AtmosphereAutopilot>); the state they act on is upstream, so ELK offers them on the official build too.
MechJeb2 2.15.3 by the MuMech team: <https://github.com/MuMech/MechJeb2>.
NavInstruments Continued (NavUtilities) 0.8.1.1 by linuxgurugamer: <https://github.com/linuxgurugamer/NavInstruments>.

All mods are reached by reflection only: nothing bundled, nothing referenced. A mod that is not installed hides its tab and turns its slots into no-ops. Every key calls the member the mod's own window or native hotkey calls. All keys ship empty. "Slot id" is the node name in the cfg file.

AtmosphereAutopilot - 26 slots

Tab **AtmosphereAutopilot**, file *ELK_AtmosphereAutopilot.cfg*. Options: **aa_autoengage** (true): a Cruise Flight or FBW key pressed while that controller is off switches master and controller on first; off = the key is a no-op until the controller is active. **aa_autoshow** (false): open the AA window when the master is switched on. **aa_vs_step** (1), **aa_alt_step** (50), **aa_hdg_step** (1): bug increments. Column "AA native key": the AA hotkey with the same effect; the tab's Import / Export / Restore buttons move bindings between ELK and AA's *Global_settings.txt*.

Manager

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
AA: Master switch	AA_MASTER_TOGGLE	TopModuleManager.Active (setter) + mainMenuGUIUpdate()	Master on/off, like AA's native master key.	master_switch_key
AA: FD (Cruise <-> FBW)	AA_FD_TOGGLE	TopModuleManager.activateAutopilot(CruiseController StandardFlyByWire)	From Cruise Flight to Standard FBW; from anything else to Cruise Flight. Switches the master on when needed.	fd_switch_key (fork)
AA: Show/hide window	AA_GUI_TOGGLE	GUIWindow.ToggleGUI() on the manager	Shows or hides AA's main window, like the native Shift+master key.	-
AA: Standard FBW	AA_SELECT_FBW	activateAutopilot(StandardFlyByWire)	Makes Standard Fly-By-Wire the active controller (master on if needed).	-
AA: Cruise Flight	AA_SELECT_CRUISE	activateAutopilot(CruiseController)	Makes Cruise Flight the active controller (master on if needed).	-

Cruise Flight (aa_autoengage applies)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
CF: Level (LVL)	AA_CF_LEVEL	CruiseController.LevelFlightMode = true	Lateral mode Level: wings level, holds the current heading.	level_flight_key
CF: Heading (HDG)	AA_CF_HEADING	CruiseController.CourseHoldMode = true	Lateral mode Heading: flies the HDG bug. AA itself refuses heading hold beyond 80 degrees of latitude.	course_hold_key
CF: Waypoint (NAV)	AA_CF_WAYPOINT	CruiseController.WaypointMode = true	Lateral mode Waypoint: navigates to the destination set in the Cruise Flight window.	waypoint_key
CF: V/S engage/disengage	AA_CF_VS_TOGGLE	vertical_control = true; height_mode = VerticalSpeed	Vertical control in V/S mode; pressed again in that mode, disengages it.	vspeed_select_key (fork)
CF: ALT engage/disengage	AA_CF_ALT_TOGGLE	vertical_control = true; height_mode = Altitude	Altitude hold; pressed again in that mode, disengages it.	altitude_select_key (fork)
CF: Vertical motion on/off	AA_CF_VERTICAL_TOGGLE	CruiseController.vertical_control	Vertical control on/off without changing its mode.	vertical_control_key
CF: ALT <-> V/S-FPA	AA_CF_SETPOINT_TYPE_TOGGLE	CruiseController.height_mode (+ prev_height_change_mode_by_hotkey)	Altitude hold to the last rate mode used (V/S or FPA) and back, like AA's native key.	toggle_vertical_setpoint_type_key
CF: Keys input mode	AA_CF_KEYS_MODE_TOGGLE	CruiseController.use_keys	Move the bugs with the stock pitch and yaw keys, on/off.	switch_key_mode
CF: V/S bug up	AA_CF_VS_UP	desired_vertsetpoint.Value += aa_vs_step	Vertical setpoint up one step (m/s in V/S mode, degrees in FPA mode).	vspeed_up_key

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
CF: V/S bug down	AA_CF_VS_DOWN	desired_vertsetpoint.Value -= aa_vs_step	Lowers the vertical setpoint by one step.	vspeed_down_key
CF: ALT bug up	AA_CF_ALT_UP	desired_altitude.Value += aa_alt_step	Raises the target altitude (ASL) by one step, in metres. Moves the bug only.	altitude_up_key
CF: ALT bug down	AA_CF_ALT_DOWN	desired_altitude.Value -= aa_alt_step	Lowers the target altitude by one step.	altitude_down_key
CF: ALT bug = current	AA_CF_ALT_SYNC	desired_altitude.Value = vessel.altitude; vertical_control = true; height_mode = Altitude	ALT bug = current ASL altitude, altitude hold engaged (never disengaged).	-
CF: HDG bug up	AA_CF_HDG_UP	desired_course.Value += aa_hdg_step (wrapped 0-360)	Turns the target heading right by one step, in degrees. Moves the bug only.	course_up_key
CF: HDG bug down	AA_CF_HDG_DOWN	desired_course.Value -= aa_hdg_step (wrapped 0-360)	Turns the target heading left by one step.	course_down_key
CF: HDG bug = current	AA_CF_HDG_SYNC	desired_course.Value = FlightGlobals.ship_heading; CourseHoldMode = true	Sets the HDG bug to the current compass heading and selects heading hold.	-

Standard Fly-By-Wire (aa_autoengage applies)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
FBW: Moderation	AA_FBW_MODERATION	StandardFlyByWire.moderation_switch	Toggles angle-of-attack and G-load moderation, like AA's native moderation key.	moderation_keycode
FBW: Rocket mode	AA_FBW_ROCKET	StandardFlyByWire.RocketMode	Toggles FBW rocket mode.	rocket_mode_keycode
FBW: Coordinated turn	AA_FBW_COORD_TURN	StandardFlyByWire.Coord_turn	Toggles coordinated turns.	coord_turn_keycode

Other modules (flags, no auto-engage)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
AA: Speed control	AA_SPEED_CONTROL	ProgradeThrustController.spd_control_enabled	Automatic speed (throttle) control on/off; effective only while a controller using it is active.	spd_control_toggle_key
AA: Thrust balancing	AA_THRUST_BALANCING	FlightModel.balance_engines	Toggles automatic thrust balancing between engines.	balancing_toggle_key

MechJeb2 - 27 slots on two tabs

Tabs **MechJeb2** and **MechJeb2+**, one file *ELK_MechJeb2.cfg*. Every key acts on the active vessel's master MechJeb on each press (no MechJeb aboard: no-op); a module locked in career is never driven. Options: **mj_autoengage** (true): a Translatron speed or Kill H/S key pressed with the Translatron off switches it on in Keep vertical first. **mj_autoshow** (false): also open the window of the module driven. **mj_trans_step** (1 m/s): speed key increment. **Copy from / to SAS**: copy the ten stock SAS bindings onto the matching SmartASS slots, or back (mapping below). The Translatron is a throttle controller with one mode at a time; the speed keys write its Speed box and live setpoint together, like the window's + / 0 / - buttons.

Tab MechJeb2: MechJeb's own part actions (Speed = 0 and Kill H/S driven directly)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
SmartASS: Off	MJ_SASS_OFF	MechJebCore.OnDeactivateSmartASSAction (SmartASS target OFF)	Disengages SmartASS attitude control.
SmartASS: Kill rotation	MJ_SASS_KILLROT	OnKillRotationAction (Target.KILLROT)	Cancels rotation and holds the current attitude.
SmartASS: Prograde	MJ_SASS_PROGRADE	OnOrbitProgradeAction (Mode.ORBITAL, Target.PROGRADE)	Points orbital prograde.
SmartASS: Retrograde	MJ_SASS_RETROGRADE	OnOrbitRetrogradeAction (Target.RETROGRADE)	Points orbital retrograde.
SmartASS: Normal+	MJ_SASS_NORMAL_PLUS	OnOrbitNormalAction (Target.NORMAL_PLUS)	Points orbit normal (+).
SmartASS: Normal-	MJ_SASS_NORMAL_MINUS	OnOrbitAntinormalAction (Target.NORMAL_MINUS)	Points orbit anti-normal (-).
SmartASS: Radial+	MJ_SASS_RADIAL_PLUS	OnOrbitRadialOutAction (Target.RADIAL_PLUS)	Points radial out, away from the body (navball radial-out marker).
SmartASS: Radial-	MJ_SASS_RADIAL_MINUS	OnOrbitRadialInAction (Target.RADIAL_MINUS)	Points radial in, towards the body.
Translatron: Off	MJ_TRANS_OFF	OnTranslatronOffAction (Translatron.SetMode(OFF))	Switches the Translatron off.
Translatron: Keep vertical	MJ_TRANS_KEEP_VERT	OnTranslatronKeepVertAction (SetMode(KEEP_VERTICAL))	Holds the vertical speed in the Speed box (0 = hover, negative = descent). Throttle only: attitude is yours unless Kill H/S is on.
Translatron: Speed = 0	MJ_TRANS_ZERO	Translatron.trans_spd.Val = 0; Thrust.TransSpdAct = 0	The window's "0" button: Speed box and setpoint to 0 in the current mode (hover in Keep vertical). Translatron off: Keep vertical first with mj_autoengage, otherwise no-op.
Translatron: Kill H/S	MJ_TRANS_KILLH_TOGGLE	Thrust.TransKillH (+ Translatron.SetMode(KEEP_VERTICAL))	Kill horizontal speed: in Keep vertical the Translatron also steers against the horizontal velocity, cancelling drift. Translatron on: toggles the flag. Off: with mj_autoengage engages Keep vertical with the flag on, otherwise toggles the flag only.
Translatron: PANIC	MJ_PANIC	OnPanicAction (Translatron.PanicSwitch())	Launch abort: throttle off, jettisons everything below the last engine stage and fires it (escape motor), 2 s full-throttle climb, then the landing autopilot lands somewhere. Pressed again: cancels.
Landing: Land somewhere	MJ_LAND_SOMEWHERE	OnLandsomewhereAction (Landing.LandUntargeted)	Starts the landing autopilot with no target.
Landing: Land at KSC	MJ_LAND_KSC	OnLandTargetAction (LandingGuidance.SetAndLandTargetKSC)	Targets the KSC pad and starts the landing autopilot on it.
Ascent: Autopilot on/off	MJ_ASCENT_TOGGLE	OnAscentAPToggleAction (Ascent.Users add/remove)	Engages or disengages the ascent autopilot (Ascent Guidance settings).

Tab MechJeb2+: what the part actions do not cover

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
SmartASS: Node	MJ_SASS_NODE	SmartASS.target = NODE; Engage()	Points along the next maneuver node. No planned node: nothing happens.
SmartASS: Target+	MJ_SASS_TARGET_PLUS	SmartASS.mode = TARGET; target = TARGET_PLUS; Engage()	Points at the selected target. No target: nothing happens.
SmartASS: Target-	MJ_SASS_TARGET_MINUS	SmartASS.mode = TARGET; target = TARGET_MINUS; Engage()	Points away from the selected target. No target: nothing happens.
Translatron: Keep surface	MJ_TRANS_KEEP_SURF	Translatron.SetMode(KEEP_SURFACE)	Holds the surface speed in the Speed box by throttle alone, pointing prograde or retrograde; never turns the vessel. Kill H/S has no effect here.
Translatron: Keep orbital	MJ_TRANS_KEEP_ORBIT	Translatron.SetMode(KEEP_ORBITAL)	Same for orbital speed.

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
Translatron: Speed +step	MJ_TRANS_SPEED_UP	trans_spd.Val += mj_trans_step; TransSpdAct = trans_spd	The window's "+" button: Speed box and setpoint up one step (m/s). Translatron off: Keep vertical first with mj_autoengage, otherwise no-op.
Translatron: Speed -step	MJ_TRANS_SPEED_DOWN	trans_spd.Val -= mj_trans_step; TransSpdAct = trans_spd	The window's "-" button: one step down. Same rules.
Landing: Land at target	MJ_LAND_TARGET	Landing.LandAtPositionTarget(LandingGuidance)	Lands on the position target set in Landing Guidance. Needs a position target and a vessel not landed.
Landing: Abort	MJ_LAND_ABORT	Landing.StopLanding()	Aborts the landing autopilot, only while it is running.
Node: Execute / abort	MJ_NODE_EXECUTE_TOGGLE	Node.ExecuteOneNode(ManeuverPlanner) / Node.Abort()	Executes the next maneuver node; pressed while running, aborts it. No node or zero delta-v: nothing.
MechJeb: Show/hide menu	MJ_MENU_TOGGLE	MechJebModuleMenu.ShowHideWindow()	Shows or hides MechJeb's main menu window.

Copy from / to SAS mapping

Stock SAS slot	SmartASS slot
SAS_STABILITY	MJ_SASS_KILLROT
SAS_PROGRADE	MJ_SASS_PROGRADE
SAS_RETROGRADE	MJ_SASS_RETROGRADE
SAS_NORMAL	MJ_SASS_NORMAL_PLUS
SAS_ANTINORMAL	MJ_SASS_NORMAL_MINUS

Stock SAS slot	SmartASS slot
SAS_RADIAL_IN	MJ_SASS_RADIAL_MINUS
SAS_RADIAL_OUT	MJ_SASS_RADIAL_PLUS
SAS_TARGET	MJ_SASS_TARGET_PLUS
SAS_ANTITARGET	MJ_SASS_TARGET_MINUS
SAS_MANEUVER	MJ_SASS_NODE

NavUtilities (NavInstruments Continued) - 1 slot

Tab **NavUtilities**, file *ELK_NavUtilities.cfg*, no options.

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
NavUtilities: HSI window	NAV_HSI_TOGGLE	NavUtilLibApp.displayHSI() + toolbar icon sync (SetTrue/SetFalse)	Opens or closes the HSI window, like a left click on the NavUtilities toolbar button (icon kept in sync).